DCP-Net: Learning Detail–Context Perception via Spatial-Frequency Guidance for Tiny Object Detection in Remote Sensing Images

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***Abstract*—Tiny object detection (TOD) in remote sensing images has long been challenging due to the low resolution of tiny objects and complex backgrounds. However, existing adaptive receptive field methods mainly focus on object and contextual information, making them susceptible to interference from complex backgrounds. This interference exacerbates the imbalance in sample assignment and hinders accurate adaptation of receptive fields to object scales, resulting in insufficient representation of fine-grained object details. In addition, most loss functions are prone to gradient instability in scenarios involving small objects. To address the aforementioned problems, this article proposes a detail–context perception network (DCP-Net), which learns detail–context perception under spatial-frequency guidance. To enhance the representation of details of tiny objects, the network is the first to leverage spatial frequency to distinguish the structural features of objects and background. By introducing a spatial-frequency guided dual-dilated convolution (SpatialFGDDConv), it effectively enhances local object details and global semantic associations while suppressing interference from complex backgrounds. Based on this convolutional unit, we further design a cross-layer fine-grained fusion module to recover lost object details at a lower computational cost. Furthermore, we introduce a Gaussian classification-regression loss to address the issues of imbalanced sample allocation and unstable gradient, while applying a decoupled probabilistic metric strategy and a multiscale enhanced decoupled head (ME-Decoupled Head) to improve localization accuracy. Extensive experimental results demonstrate the effectiveness of the proposed method. Specifically, DCP-Net achieves an average precision (AP) of 27.5% on the AI-TODv2 dataset and an AP50 of 78.0% on the DOTA-v1.0 dataset. The code will be found at: https://github.com/yangyvni/DCP-Net**

***Index Terms*—Multiscale fusion, remote sensing images, spatial-frequency guidance, tiny object detection (TOD).**

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I. Introduction

TINY object detection in remote sensing images is a core task in remote sensing image processing, aiming to classify and localize objects of interest. It plays a critical role in applications such as search and rescue [1], military reconnaissance [2], and traffic monitoring [3]. However, remote sensing images usually cover large areas and contain numerous small-scale objects with diverse shapes and high density. These objects occupy a few pixels and exhibit weak features. In complex scenes with occlusion and illumination variation, object boundaries become blurred, causing label assignment to favor background regions. Consequently, tiny object detection (TOD) in remote sensing imagery remains highly challenging [4].

In recent years, most studies have adopted spatial-domain methods based on multiscale feature fusion to enhance tiny object representations [5], [6], [7]. By integrating deep semantic features with shallow high-resolution features, these methods alleviate the loss of tiny object details to some extent. However, they still mainly rely on deep standard convolutional networks to recover local features, while lacking global relationship modeling. Such indiscriminate feature fusion easily introduces background noise and semantic redundancy in complex scenes.

To address the limitations of local feature modeling and complex background interference, some methods introduce adaptive receptive field mechanisms to enhance feature sampling for objects of different scales under complex scenes. These methods improve geometric adaptability through offset learning [8], expand semantic context via dynamic dilation strategies [9], [10], [11], or suppress background interference through attention-guided feature enhancement [12]. Although such methods alleviate the limitations of fixed receptive fields in standard convolutions, their receptive field ranges remain difficult to control effectively. In densely distributed tiny object scenarios, excessively enlarged receptive fields may skip tiny objects and lead to discontinuous sampling, making it difficult to distinguish object features from complex backgrounds.

In addition, some recent methods [13], [14], [15] have shifted from the spatial domain to the frequency domain to enhance object-related frequency responses by amplifying high-frequency object components while suppressing low-frequency background information. However, repeated spatial-frequency transformations substantially increase

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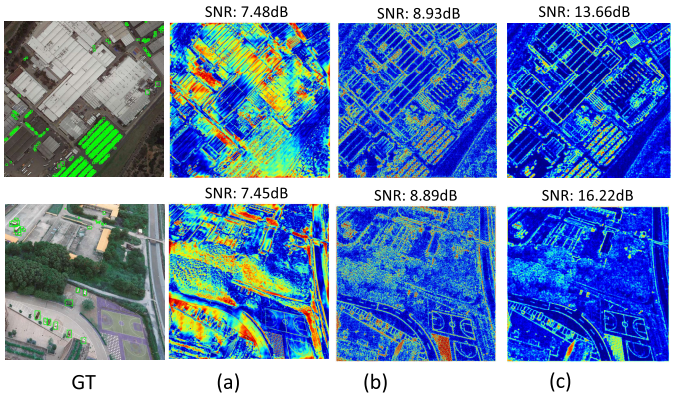


Fig. 1. Heatmap visualization of the high-frequency features [including signal-to-noise ratio (SNR)] in different domain enhancement methods (spatial and frequency domains) on the image of the AI-TODv2 dataset. (a) FFT [13]: foreground–background features are severely confused. (b) FADConv(FreqSelect) [14]: reduced the background attention significantly, but noise remains. (c) Ours(HighFreqProxy): significantly higher SNR with clearer foreground–background separation.

inference latency, and the resulting global receptive fields lack awareness of spatial structural distributions. Consequently, background regions with similar frequency characteristics are easily amplified together with target responses, leading to increased noise interference, as illustrated in Fig. 1.

To effectively guide the training of deep models, existing methods have extensively explored loss functions for object detection. Intersection over union (IoU)-based losses [16], [17], [18] are overly sensitive to positional and shape variations between predicted boxes and ground-truth (GT) boxes, making them prone to gradient vanishing for tiny objects. Although the normalized Wasserstein distance (NWD) [19] alleviates this issue by measuring distribution similarity from a probabilistic perspective, these methods still represent regression deviations using a single scalar metric, leading to strong parameter coupling and unstable gradient optimization among position, shape, and scale regression tasks, especially in densely distributed tiny object scenarios.

Therefore, this article proposes a learning detail–context perception network (DCP-Net) via spatial-frequency guidance, designed for TOD under a one-stage framework, aiming to address TOD from both feature representation and training optimization perspectives. First, we design a spatial-frequency-guided dual-dilated convolution (SpatialFGDDConv), which consists of three core components.

1) A lightweight squeeze-and-excitation attention mechanism (LiteSEAttention) to select effective channel features.
2) A high-frequency proxy module (HighFreqProxy) that focuses on capturing fine edge details of small objects.
3) An adaptive dual-dilated convolution dominance module (ADualDCDM) to overcome the limited flexibility of fixed receptive fields in feature extraction.

Compared with existing dynamic dilation strategies, this scheme significantly enhances the feature discrimination ability between tiny objects and background under the guidance of spatial frequency in HighFreqProxy, thereby controlling ADualDCDM to extract appropriate receptive fields. This structure enables the model to flexibly perceive object details, contextual textures, and background information through the collaborative effect of these three modules, thereby selectively enhancing object features in deep networks, while fully associating global context to suppress background interference. Moreover, by leveraging the interference suppression capability of SpatialFGDDConv, we propose a cross-layer fine-grained Fusion enhancement (CFFE) structure. This structure deeply integrates the shallowest high-resolution features across layers, which not only recovers lost object details but also achieves purer multiscale fusion with minimal computational cost.

Second, we propose a Gaussian classification–regression loss (GCls-RegLoss) to supervise the spatial-frequency guidance network. Since the guidance process emphasizes high-frequency edge responses, it is susceptible to background edge interference. By introducing Gaussian-based probability estimation, the loss assigns larger weights to hard samples for more accurate foreground–background discrimination. Meanwhile, it decouples position and shape optimization to stabilize regression gradients and introduces an attraction-repulsion mechanism to reduce interference in dense scenarios. Furthermore, we design a multiscale enhanced decoupled head (ME-Decoupled Head), which captures geometric structures of tiny objects through parallel convolutions with different kernel sizes, thereby providing more accurate spatial matching for regression optimization.

Our main contributions are summarized as follows.

1) To alleviate the loss of local details and the insufficient global feature relationships in deep convolutional networks, we propose a SpatialFGDDConv. Through the synergistic effects of HighFreqProxy, ADualDCDM, and LiteSEAttention, this convolution structure uses a HighFreqProxy map in the spatial domain to capture the objects' edge texture, guiding the model to achieve an adaptive balance between preserving high-frequency detail and capturing global context. By the SpatialFGDDConv, we propose a CFFE structure to supplement the spatial details of high-level features further.
2) We introduce GCls-RegLoss, which not only solves the problems of sample class assignment and existing loss, but also can competitively handle object-dense scenarios. To obtain more accurate prediction results, a ME-Decoupled Head is designed to enhance object details while compensating for surrounding spatial context features, helping the model distinguish objects of different scales and shapes from complex backgrounds, and improving localization accuracy.
3) Extensive experiments on two datasets demonstrate the effectiveness of the proposed method. Compared to state-of-the-art (SOTA) methods, it achieves significant performance improvements, particularly when handling very tiny and tiny objects.

The remainder of this article is organized as follows. Section II reviews relevant works. Section III details the proposed methods. Section IV presents the experimental results and settings. Finally, Section V summarizes this article.

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TABLE I

STUDY ON TYPICAL SMALL OBJECT DETECTION METHODS. AR, PARAMS, GFLOPS, AND FPS DENOTE AVERAGE RECALL, PARAMETER SIZE, GIGA FLOATING POINT OPERATIONS, AND FRAMES PER SECOND. VALIDATION METRIC RESULTS ON DIFFERENT DATASETS ARE SEPARATED BY SEMICOLONS

Method	Dataset	Metric Result	Mechanism & Effect (Pros)	Limitations (Cons)
MPFPN [5]	VisDrone-DET	AP: 29.05 AR: 45.69	Recovers small-scale features by introducing extra branches for deep layers.	Neglects global context; prone to background aliasing.
HyperNet [6]	PASCAL VOC2007	AP_{50}: 76.3 AR: 97.0	Optimizes small object representation via deep-shallow feature integration.	High computational redundancy.
SCDNet [33]	DTOD	AP_{50}: 24.2	Captures global spatial dependencies through scene context guided fusion module (SCGM).	Lacks local adaptability, thereby losing tiny objects' details when integrating the global context.
MFEL-YOLO [8]	VisDrone 2019; AI-TOD	AP: 27.3; AP: 15.3 AR_{vt}: 13.5	Enhances spatial feature extraction using deformable convolution v4 (DCNv4)-based adaptive extraction unit (AEU).	High complexity due to offset learning.
SED-YOLO [10]	DOTA-v1.0	AP_{50}: 71.6 Params: 8.09M FPS: 250	Balances local/global features via Switchable Atrous Convolution (SAConv) with attention.	Sensitive to pre-defined dilation ranges; Attention may misactivate in complex backgrounds.
ADC [11]	VisDrone 2019	AP: 38.88 AP_{50}: 59.12 AP_s: 28.37	Frequency-based adaptive dilation convolution (ADC) enables dual enhancement of details and context.	Reliance on frequency partitioning leads to poor generalization; High training cost due to frequency domain transformation.
SEMA-YOLO [12]	RS-STOD; AI-TOD	AP_{50}: 72.5 Params: 3.6M GFLOPs: 14.2 FPS: 185; AP_{50}: 61.5	Captures precisely small object details through multi-scale adaptive attention.	Overlapping receptive fields in dense scenes.
RFAG-YOLO [34]	VisDrone 2019	AP: 23.1 Params: 5.94M GFLOPs: 15.7 FPS: 82	Boosts local fine-grained extraction by receptive field network (RFN) blocks with inverse residual paths.	Purely local weighting fails to discern complex backgrounds globally, causing receptive field misactivation.
Position-DETR [35]	DOTA-v1.0; DIOR; VisDrone 2019	AP: 42.3; AP: 65.5; AP: 25.3	Enhance small object spatial features and boost the model's convergence speed via tailored losses.	Excessive hyperparameter tuning (5 parameters).
NWD [19]	AI-TOD; VisDrone 2019	AP: 17.8 (on [22]); AP_{50}: 38.5 (on [22]) AP_{vt}: 3.8	Solves IoU sensitivity to tiny shifts using Wasserstein distance metric.	Extreme aspect ratios can exacerbate parameter coupling under a single scalar metric, leading to unstable regression gradients.
DotD [36]	AI-TOD	AP: 14.9 AP_{vt}: 7.2 (on [22])	Simple computation; resolves gradient vanishing via normalized center Euclidean distance.	Lacks aspect ratio/scale modeling, causing unstable gradients.

II. RELATED WORK

A. Small Object Detection

Existing detectors are usually divided into two types: convolution-based and transformer-based. Convolution-based detectors can be further divided into two-stage detectors represented by the R-CNN series [20], [21], [22], [23], [24] and one-stage detectors represented by the YOLO series [25], [26], [27], [28], [29], [30]. Transformer-based detectors are mainly represented by DETR [31], [32]. However, general object detectors usually perform well on objects of regular scales but are limited in detecting small objects.

Most small object detection methods are based on general detectors and are improved through feature extraction, fusion, and loss functions, as summarized in Table I. Existing feature fusion [5], [6], [33] tends to cause information redundancy or loss, making it difficult to balance local and global key features. Although adaptive receptive field methods [8], [10], [11], [12], [34] can dynamically extract multiscale features, they rely on predefined rules or complex modeling and are susceptible to interference from complex backgrounds, making it difficult to achieve precise and adaptive adjustment in densely distributed small object scenarios.

The proposed method does not rely on predefined adjustment logic. It leverages the intrinsic frequency properties of images to distinguish objects directly from background, thereby adaptively fusing local and global receptive field features at the pixel level. Combined with efficient multiscale cross-layer fusion, it addresses these limitations. Additionally, we alleviate the problems of the complex parameter tuning and unstable gradient localization in existing loss functions [19], [35], [36] to ensure accurate and stable training.

B. Frequency Sensing and Enhancement in Object Detection

Due to the limitations of spatial-domain methods, frequency analysis has been introduced into object detection research, especially in the field of remote sensing images. Existing works mainly employ the discrete Fourier transform (DFT) [14], [37], [38] or the discrete cosine transform (DCT) [15] to extract high-frequency components for feature enhancement. For example, [14] proposed a frequency selection (FreqSelect) module, which performs multiband decomposition using FFT, and applies spatially adaptive weighting to preserve object details and suppress background high-frequency components. In recent years, frequency-domain enhancement methods specifically designed for small object detection [37], [38] have emerged. For instance, Wen et al. [37] propose a multiscale frequency feature enhancement module (MSFFEM) that adaptively enhances the contours and texture features of small objects using block-wise FFT, and [38] incorporates frequency-domain processing based on the CSP network to effectively capture global contextual features and enhance the distinction between small objects and backgrounds. In addition, [15] applies high-pass filtering to extract high-frequency responses in FPN as spatial and channel mask weights to highlight tiny objects.

However, these methods may introduce the Gibbs phenomenon, leading to blurred object boundaries, and global responses can introduce irrelevant background frequency interference, which is unfavorable for small object enhancement. Meanwhile, the transformation process significantly increases inference latency. Some studies also adopt the wavelet transform (WT) [39], which maps features into subband space without loss through wavelet decomposition for enhancement. Although this alleviates the Gibbs effect, it increases channel redundancy.

To avoid these limitations, we approximate frequency responses directly in the spatial domain to emphasize the local high-frequency structure of objects. The Laplacian convolution employed uses second-order differentiation, which strongly responds to "isolated high-frequency" objects while suppressing "continuous high-frequency" background regions. This design achieves effective discrimination between objects and background at low computational cost, leading to improved detection performance.

C. Label Assignment Strategy in Object Detection

In object detection tasks, label assignment strategies [22], [23], [40], [41] directly govern model performance, whose primary objective is to categorize samples into positive samples (those matching GT labels) and negative samples (those not

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matching GT labels). Usually, sample categorization is performed based on the IoU metric: a predicted box is designated as a positive sample if its IoU with the corresponding GT box exceeds a predefined threshold, and a negative sample otherwise. Traditional hard label assignment strategies [22], [23], [42], [43] employ fixed rules to clearly distinguish positive and negative samples. However, a large number of easy and negative samples tend to dominate the model's gradient update, neglecting the learning of hard samples, which significantly degrades the model's detection accuracy. To divide the samples more flexibly, high-precision detectors employ soft label assignment strategies [44], [45]. NWD-RKA [44] allocates training samples according to the ranking results of the Gaussian Wasserstein distance (GWD) [46] scores between each anchor box and the GT box. The Ali-TSM strategy [45] dynamically demarcates the boundary between positive and negative samples based on predictive information, balancing the number of positive samples for objects across different scales. Nevertheless, these strategies require extensive manual hyperparameter tuning and exhibit poor generalization ability to diverse sample datasets.

Therefore, we adopt a hybrid label assignment strategy. Instead of fixed thresholds, it dynamically selects positive samples based on both localization and classification quality, and assigns adaptive weights using Gaussian statistics. This strategy mitigates class imbalance, removes manual hyperparameter tuning, and improves training robustness.

III. Methodology

This section first introduces the overall architecture of the DCP-Net in detail. and then describes the SpatialFGDDConv and its application to the CFFE structure. Finally, the loss function GCls-RegLoss, including the design of the ME-Decoupled Head, is discussed.

A. Overall Architecture

The overall architecture of DCP-Net is shown in Fig. 2. It consists of a SpatialFGDDConv, CFFE, and ME-Decoupled Head, and is optimized by GCls-RegLoss. Among them, SpatialFGDDConv extracts object details from the input image while suppressing background interference, and is combined with BottleNeck and SPPF in YOLO11 [30] to further enhance the model's perception of local features and global semantics. The CFFE structure introduces the P2 layer across scales to optimize gradient propagation between shallow and deep features, and incorporates SpatialFGDDConv to achieve selective multiscale feature fusion, thereby improving receptive field fusion capability. The ME-Decoupled Head receives the fused features from CFFE to achieve more accurate localization for multiscale and multishape objects by reconstructing the regression branch, ensuring that GCls-RegLoss can more stably supervise network optimization and improve overall detection performance.

B. Spatial-Frequency-Guided Dual-Dilated Convolution

To accurately extract weak small-scale object features from complex backgrounds, we propose a novel convolutional unit, namely the SpatialFGDDConv, as shown in Fig. 3. This unit consists of three parts: LiteSEAttention, HighFreqProxy, and ADualDCDM.

First, the input feature X_i passes through a LiteSEAttention mechanism. This mechanism prioritizes filtering channel-level features to reduce redundancy for subsequent frequency analysis. It also strengthens object-related feature channels, such as edge channels, and then outputs feature X_o.

The specific is formulated as

$$\begin{aligned} X' &= AvgPool\left(X_i\right) \\ X_o &= X_i \times Sigmoid\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}\left(X'\right)\right)\right)\right) \end{aligned} \tag{1}$$

where $X_i \in R^{C_i\times H\times W}$ (C_i is the number of input channels, $H \times W$ is the input size) are the input features, $X' \in R^{C_i\times1\times1}$ are the channel-compressed features, and $X_o \in R^{C_i\times H\times W}$ are the clean features after channel filtering.

Next, we designed the HighFreqProxy module, which uses lightweight Laplacian convolution to approximate the local frequency intensity of each pixel in X_o. After that, the mean filtering is performed to smooth the output signal and suppress noise, generating a normalized HighFreqProxy map that provides a stable spatial guiding signal $W_s \in R^{1\times H\times W}$, which helps to locate high-frequency foreground objects and low-frequency background interference accurately. HighFreqProxy can be described by the following equation:

$$W_s = Sigmoid\left(Filter\left(Conv_{lap}\left(X_o\right)\right)\right) \tag{2}$$

where $Conv_{lap}(\cdot)$ represents the Laplacian convolution operation, and $Filter(\cdot)$ denotes the mean filtering operation.

At the same time, we utilize the LiteSEAttention mechanism to calculate the channel attention map and generate channel guidance signals $W_c \in R^{C_o\times H\times W}$ (where C_o is the number of output channels). To comprehensively understand high-frequency foreground and low-frequency background, we fuse spatial position weights W_s and channel semantic weights W_c. The high-frequency attention weight $W_h \in R^{C_o\times H\times W}$ is computed as $W_h = W_c \times W_s$ and is used to enhance the high-frequency object region. The low-frequency attention weight $W_l \in R^{C_o\times H\times W}$ is computed as $W_l = (E - W_s) \times (E - W_c)$ and is used to unify the low-frequency background region.

Then, we input X_o into the ADualDCDM, which consists of two parallel dilated convolutions, representing the high-frequency branch and low-frequency branch, respectively. The specific implementation formula of ADualDCDM is as follows:

$$Y_1 = HighFreqConv_{rate=1}\left(X_o\right) \tag{3}$$

$$Y_2 = LowFreqConv_{rate=2}\left(X_o\right) \tag{4}$$

$$Y = W_h \times Y_1 + W_l \times Y_2 \tag{5}$$

where $HighFreqConv_{rate=1}(\cdot)$ and $LowFreqConv_{rate=2}(\cdot)$ represent dilated convolution operations with dilation rates of 1 and 2, respectively.

Guided by W_h and W_l, the clean feature X_o dynamically adjusts the receptive field allocation strategy according to spatial frequency characteristics. In high-frequency regions, a smaller dilation rate is used to preserve fine edge details of small objects, yielding $Y_1 \in R^{C_o\times H\times W}$. In low-frequency regions, a larger dilation rate is used to capture long-range contextual relationships, yielding $Y_2 \in R^{C_o\times H\times W}$, and this

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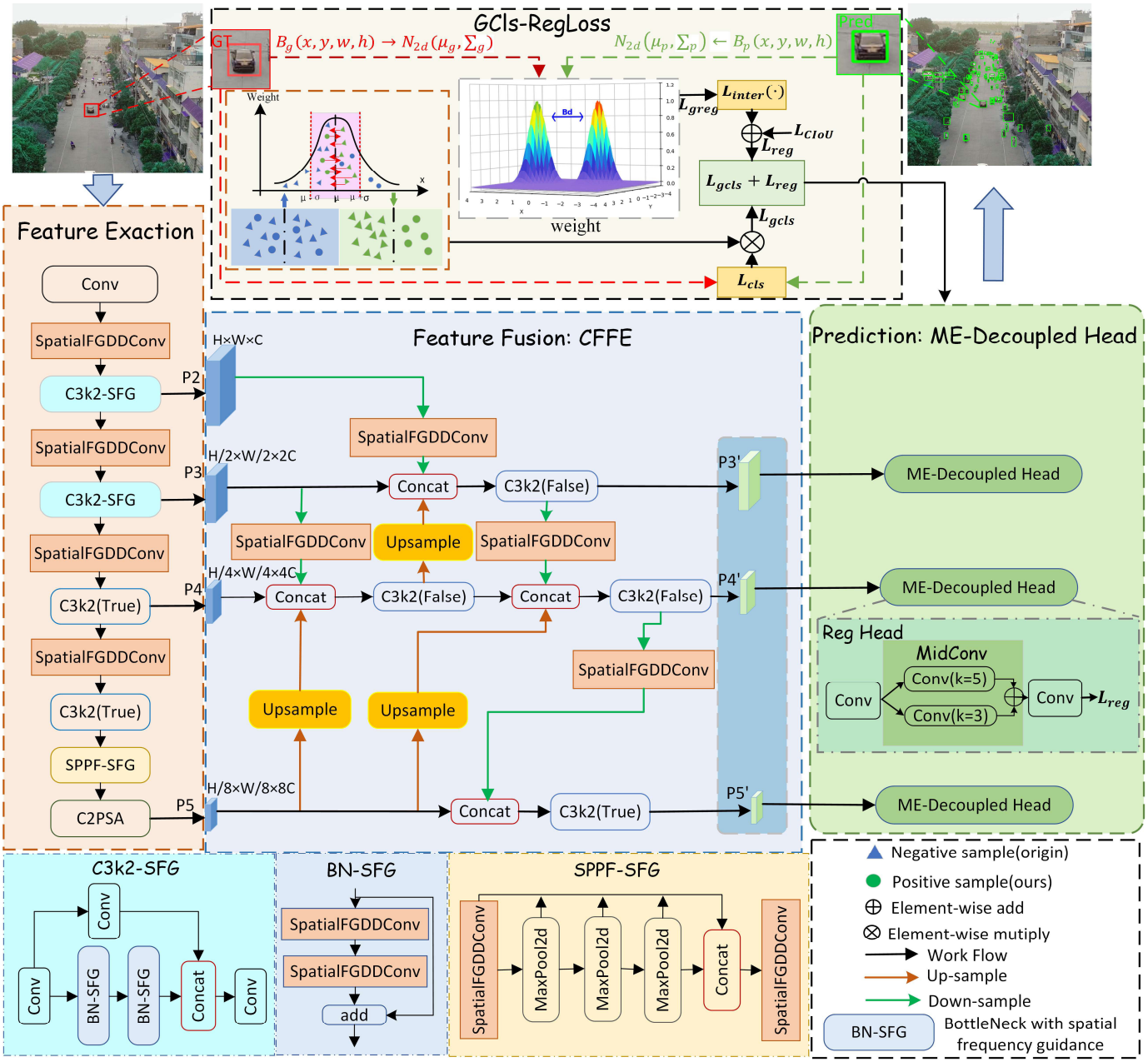


Fig. 2. Overall structure of DCP-Net. It comprises three stages: feature extraction, feature fusion, and the prediction head. SpatialFGDDConv is utilized as the first two stages to obtain more detailed and richer contextual features, which are further enhanced by the CFFE structure and passed to the prediction head for classification and regression. GCls-RegLoss is used to supervise this propagation process.

global semantic consistency is used to suppress low-frequency background interference. In addition, this large receptive field design suppresses isolated high-frequency background noise through regional feature aggregation.

Finally, by fusing the outputs of the two branches, the model achieves joint learning of foreground detail enhancement and background suppression, improving object–background contrast and producing enhanced features $Y \in R^{C_o \times H \times W}$. SpatialFGDDConv combines spatial-frequency guidance with dual-dilated convolution, effectively addressing the loss of object features in complex backgrounds, and is particularly suitable for small object detection.

To further validate the design theories of SpatialFGDDConv, we conducted a step-by-step visualization experiment. We extracted intermediate features from each stage and visualized their heatmaps on AI-TODv2 images. As shown in Fig. 4, LiteSEAttention can suppress some redundant noise (such as vegetation and buildings). Additionally, HighFreqProxy highlights the edges of small objects (such as vehicles and pedestrians), making the model focus more on high-frequency object regions. The inverse W_l map focuses more on low-frequency background regions. Meanwhile, in ADualDCDM, feature maps with a small dilation rate have a higher response to small-scale objects with high-frequency details, while

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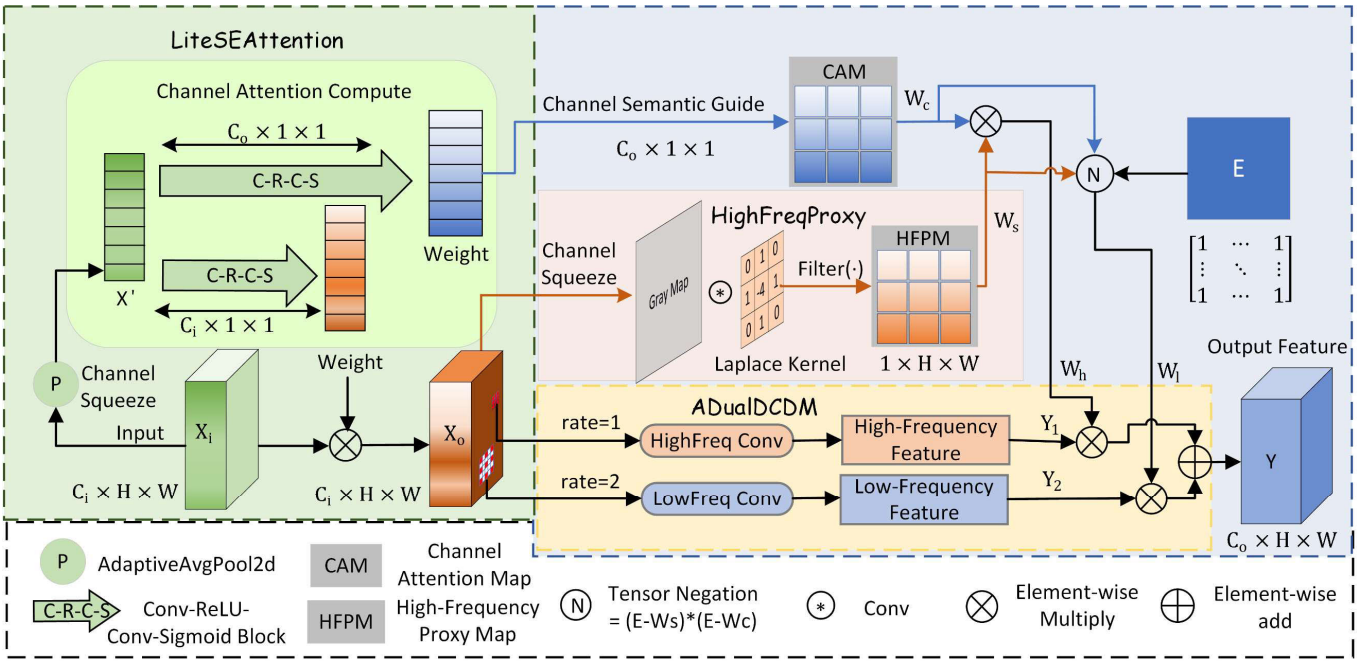


Fig. 3. Structure of SpatialFGDDConv, including three components: LiteSEAttention, HighFreqProxy, and ADualDCDM.

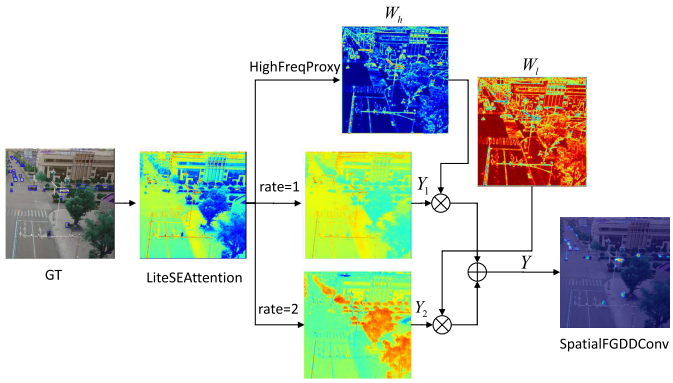


Fig. 4. Heatmap display of intermediate features at each stage of the SpatialFGDDConv. The red, yellow, green, and blue areas represent the decreasing strength of the network response in that order.

feature maps with a large dilation rate have a higher response to global background structure with low-frequency contextual information and smooth some high-frequency background interference.

Finally, by adaptively fusing the high-frequency and low-frequency features, contextual information can be effectively utilized, and background can be suppressed, thereby precisely focusing attention on the objects of interest.

In CNN, as the network depth increases, the resolution of the output feature maps decreases. Although the semantic information becomes more abstract, a large amount of detailed information about the object is lost, especially for small-scale objects. Therefore, the shallow features have high resolution and can preserve details such as edges and textures, which help locate small objects. However, the conventional FPN fusion strategy only recovered the little lost features through the two-layer concat operation of "indiscriminate stacking," and did not fully utilize the information of different layers (especially the shallow layer), resulting in the loss of information during both top-down and bottom-up propagation, weak perception ability of small objects, and synchronous fusion of background noise. To address these problems, we propose a CFFE structure by taking advantage of SpatialFGDDConv's ability to enhance small object features and suppress noise interference, as shown in Fig. 1. Downsampling uses SpatialFGDDConv for effective feature selection, and upsampling adopts bilinear interpolation to alleviate the problem of feature discontinuity. We achieve bidirectional scale fusion through cross-layer design. In addition, this structure fuses the fine-grained features from the shallowest layer (P2), enhancing the model's feature representations of small objects. This cross-level gradient flow enables the deep gradients to directly backpropagate to the P2 layer, optimizing the extraction of shallow features.

C. Gaussian Classification-Regression Loss

To alleviate blurred object boundaries and unstable localization, we propose a GCls-RegLoss, defined as Loss in (17), which consists of Gaussian classification loss (GClsLoss) and regression loss (RegLoss). GClsLoss addresses imbalanced sample assignment, while the Gaussian regression loss (GRegLoss) in RegLoss mitigates gradient vanishing in IoU-based optimization for small objects and improves the gradient instability problem caused by the coupling of metric parameters. Both losses use 1-D or 2-D Gaussian distributions to model the sample weights and regression boxes, respectively. Specifically, for classification loss, the threshold for dividing positive and negative samples is adaptively adjusted according to the IoU quality score Q_i between each sample and its ground truth. Samples near the threshold are assigned higher weights, encouraging the model to focus on hard examples and improving training efficiency. As shown in Fig. 2, the mean

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IoU quality score is used as the Gaussian mean μ, and the variance σ is estimated accordingly. The probability weight of each sample is obtained according to (8), and finally weighted into the corresponding category loss to obtain the GClsLoss function $Loss_{gcls}$. In this way, positive samples, especially hard samples, can obtain a larger loss gradient, thereby balancing the distribution of positive and negative samples. This helps to clarify the boundaries between the object and the background and improves the accuracy and robustness of classification in complex scenes

$$Q_i = Cls_score \times IoU_{pos}(pred, GT) \tag{6}$$

$$weight = P\left(x \mid \mu_k, \sigma_k^2\right) + 1 \tag{7}$$

$$= \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right) + 1 \tag{8}$$

$$\mu_k = \frac{\sum_k Q_i}{Pos_num} \tag{9}$$

$$Loss_{gcls} = (BCELoss + FocalLoss) \times weight \tag{10}$$

where Cls_score represents the confidence score of the GT box category corresponding to the predicted box in the positive sample, $IoU_{pos}(pred, GT)$ represents the IoU between this predicted box and the GT box, Q_i represents the IoU quality score between the ith positive sample and the GT box, Pos_num represents the total number of positive samples, and $P(x \mid \mu_k, \sigma_k^2)$ represents that the sample x follows a Gaussian distribution of $N(x \mid \mu_k, \sigma_k^2)$, where k refers to the sample batch.

For regression loss, we transform the problem of IoU overlap between two bounding boxes into a problem of similarity between two distributions. Considering the statistical characteristics of the Gaussian distribution, which is "dense at the center and decaying at the edges," its probability distribution-based soft positioning index method is better able to describe the spatial location of small objects with weak features and ambiguous boundaries than the geometric coordinate-based hard positioning index. First, we model the bounding box coordinates (x, y, w, h) as a 2-D Gaussian distribution $N(\mu, \Sigma)$. To match the probability range of the Gaussian distribution with the physical size of the bounding box, we calculate the variance parameter Σ using a uniform distribution, as shown in the following equation:

$$\mu_p = \left(x_p, y_p\right)^T, \quad \mu_g = \left(x_g, y_g\right)^T$$

$$\Sigma_p = \begin{bmatrix} a_p & 0 \\ 0 & b_p \end{bmatrix} \quad \Sigma_g = \begin{bmatrix} a_g & 0 \\ 0 & b_g \end{bmatrix}$$

$$a_i = \frac{w_i^2}{12}, \quad b_i = \frac{h_i^2}{12} \quad (i = p, g) \tag{11}$$

where the subscripts p and g represent the predicted box and the GT box, respectively, (x_i, y_i, w_i, h_i) represent the center coordinates (x_i, y_i), width, and height of the bounding box, respectively, and μ, Σ represent the mean vector and covariance matrix of the Gaussian distribution

$$f(x, y) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\right) \tag{12}$$

where the coordinates $z = (x, y)^T$ of the bounding box satisfy the 2-D Gaussian probability density function $f(x, y)$.

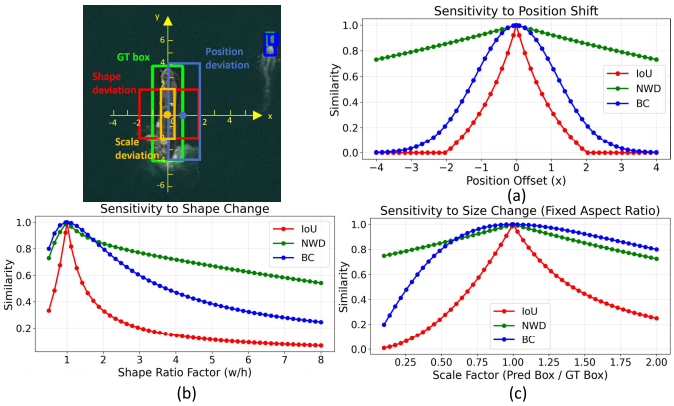


Fig. 5. (a)–(c) Sensitivity comparisons of BC, IoU, and NWD to positions, shape, and scale deviations. The comparison results are based on the settings in the top image. The results show that IoU is prone to gradient vanishing in the detection bounding boxes of small objects, and it is clear that both BC and NWD methods alleviate the gradient vanishing problem. However, the NWD method is the least sensitive to these three types of offsets, resulting in slow convergence and insufficient optimization.

Then, the similarity BC between the two Gaussian distributions is measured based on the Bhattacharyya distance [47] B_d, and finally mapped to the Gaussian regression loss $Loss_{greg}$

$$B_1 = \frac{1}{4}\left(\frac{\Delta\mu_x^2}{a_p + a_g} + \frac{\Delta\mu_y^2}{b_p + b_g}\right)$$

$$\Delta\mu_x^2 = \left(x_p - x_g\right)^2, \quad \Delta\mu_y^2 = \left(y_p - y_g\right)^2$$

$$B_2 = \frac{1}{2}\ln\left(\frac{\det\left(\Sigma_p + \Sigma_g\right)}{4\sqrt{a_p b_p a_g b_g}}\right)$$

$$B_d = B_1 + B_2$$

$$BC(p, g) = \exp(-B_d) \tag{13}$$

where $\det(\Sigma_p + \Sigma_g) = (a_p + a_g)(b_p + b_g)$, B_1 represents the center offset term, which measures the impact of the center offset of the two bounding boxes on the overlap of the probability distribution, and B_2 represents the shape difference term, which measures the impact of the aspect ratio and size difference of the two bounding boxes on the overlap of the probability distribution

$$Loss_{greg}(p, g) = \sqrt{1 - BC(p, g)}. \tag{14}$$

It is evident from (13) and (14) that the similarity BC and the GRegLoss function $Loss_{greg}$ are both greater than 0, and the gradient is continuous. This fundamentally solves the gradient vanishing problem caused by the small IoU of small objects. Moreover, by jointly quantizing the differences in center and shape, adaptive adjustment of the center and shape offset of the predicted box for small objects is achieved.

We plotted the similarity curves for the three measures of IoU, NWD, and BC as position, shape, and scale changes. As shown in Fig. 5, our BC measure method achieves a better balance between robustness and sensitivity, not only solving the gradient vanishing problem of IoU but also improving the perception of position, shape, and scale bias compared to NWD. Although NWD alleviates the gradient vanishing issue, it relies on a single distribution distance to measure similarity. Center deviation and shape discrepancy are implicitly coupled,

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making it difficult for the model to optimize for variations in position or shape, which are critical for tiny object localization. Different from the NWD metric, the BC metric splits the "position difference" and "shape (including scale) difference" into two independent terms through B_1 and B_2 terms, and assigns greater losses to the objects with significant differences, respectively, so that the model can preferentially correct the positioning errors of these objects, thereby providing a more stable and effective regression gradient.

Because remote sensing images contain a large number of dense small objects with severely overlapping bounding boxes, they are easily disturbed by surrounding objects during the regression process, causing the predicted boxes to "drift" and reducing localization accuracy. We combine the similarity BC calculated by Gaussian regression loss to determine the "attraction" between the predicted box and the GT box. Meanwhile, a piecewise function assigns higher "repulsion" loss to low-similarity boxes, reducing over-dense predictions and low-quality matches in small object scenarios. This "attraction–repulsion" mechanism specifically addresses occlusion interference of dense small objects, enhancing localization stability and enabling predicted boxes to more accurately regress to the corresponding GT boxes. The detailed computation is shown in (15)

$$\begin{aligned} Loss_{inter} &= \alpha L_{Apg} + L_{Rpg} + L_{Rpp} \qquad (15)\\ L_{Apg} &= Loss_{greg}\left(B_i, GT_i\right)\\ L_{Rpg} &= smooth\left(BC\left(B_i, GT_j\right)\right)\\ L_{Rpp} &= smooth\left(IoU\left(B_i, B_j\right)\right) \qquad (16) \end{aligned}$$

where L_{Apg}, L_{Rpg}, and L_{Rpp} represent the attraction loss between the predicted box and its own GT box, the repulsion loss between the predicted box and its surrounding GT box, and the repulsion loss between the predicted box and its surrounding predicted boxes, respectively. B_i represents the ith predicted bounding box, GT_j represents the jth GT bounding box, and smooth is a piecewise function that adjusts the repulsion loss weights. The final regression loss is $Loss_{reg} = a \times CIoULoss + Loss_{inter}$ (where $a + \alpha = 1$).

The total loss function is expressed as follows:

$$Loss = Loss_{gcls} + Loss_{reg}. \qquad (17)$$

The synergistic effect of the two loss functions can reasonably distinguish positive object samples and negative background samples and stabilize the gradient information used for localization.

Although the proposed regression loss can provide a stable gradient signal, its effectiveness depends on whether the detection head can fully perceive the specific location of the object. The decoupled detection head is an optimized version of the detection head in one-stage detectors, which addresses the issue of feature interference between classification and regression tasks in the coupled detection head. However, its regression branch still uses a single receptive field, struggling to capture the local spatial features (such as edges and shapes) of small objects at different scales. Therefore, we propose an ME-Decoupled Head, as shown in Fig. 2. It adds parallel convolutional branches with different kernel sizes (3×3 and 5×5) between two convolutional kernels to cover different receptive fields, allowing us to capture the fine edges of small objects and the spatial context information surrounding the object separately. After fusing the parallel branches, the output features can form more discriminative multiscale geometric representations, helping GClsRegLoss more precisely optimize the prediction results of it.

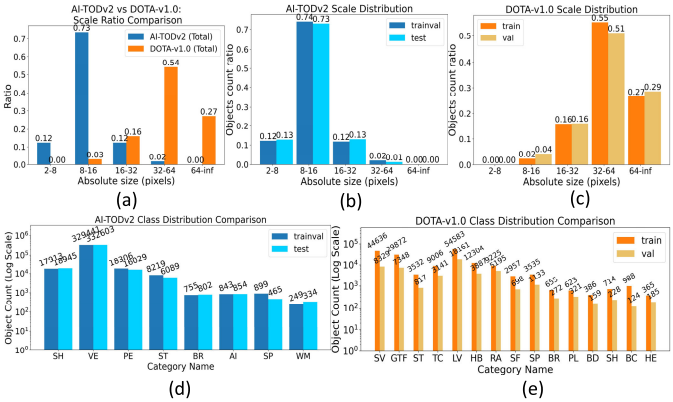


Fig. 6. Object characteristic distributions of the AI-TODv2 and DOTA-v1.0 datasets and their train/validation subsets. (a) Scale distribution comparison on both datasets. (b) and (d) Scale and category distributions of the trainval/test subsets in AI-TODv2. (c) and (e) Scale and category distributions of the train/val subsets in DOTA-v1.0.

TABLE II

TRAINING SETUP FOR DIFFERENT DATASETS. I-S DENOTES IMAGE SIZE. LR AND BS DENOTE LEARNING RATE AND BATCH SIZE. E AND EM INDICATE EPOCHS AND EVALUATION METRICS, RESPECTIVELY. * DENOTES THE OPTIMAL SETTINGS FOR THE MODEL TO REACH ITS OPTIMAL STATE

Datasets	I-S	Train:Test Count	LR	BS	E	EM
AI-TODv2 [44]	640^2	14018:14018	0.01	8/4*	300/700*	AP,GF,PA,FPS
DOTA-v1.0 [48]	640^2	12682:3909	0.01	4	500	AP50,GF,PA,FPS

IV. EXPERIMENTS

A. Datasets

1) AI-TODv2 [44]: This dataset is specifically designed for small-scale object detection in aerial images, containing 28 036 images and 752 745 object instances across eight categories. About 86% of objects are smaller than 16 pixels, the maximum size is less than 64 pixels, and the average size is 12.7 pixels. We follow the official split, using the trainval set with 14 018 images for training and the test set with 14 018 images for testing. The object distributions (scale and category) of each subset are shown in Fig. 6(b) and (d).

2) DOTA-v1.0 [48]: This dataset is a remote sensing dataset composed of satellite and aerial images, providing abundant and challenging small object samples. It contains 2806 images and 188 282 instances, including 15 categories. To reduce preprocessing loss, we use the official image-splitting tool to crop images into 800×800 patches with 200-pixel overlap, generating 12 682 training images and 3909 validation images. The object distributions (scale and category) are shown in Fig. 6(c) and (e).

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TABLE III

PERFORMANCE COMPARISON OF DCP-NET AND OTHER SOTA METHODS ON THE AI-TODV2 TEST SET. "W/" DENOTES WITH A SPECIFIC MODULE OR METHOD

Method	Venue	Backbone	AP	AP50	AP75	APvt	APt	APs	APm
two-stages detectors:									
Faster R-CNN [22]	TPAMI 2017	ResNet-50	11.1	26.3	7.6	0.0	7.2	23.3	33.6
Cascade R-CNN [23]	CVPR 2018	ResNet-50	13.8	30.8	10.5	0.0	10.5	25.5	36.6
Cascade R-CNN w/NWD [19]	ISPRS 2021	ResNet-50	18.7	44.2	12.9	3.6	17.4	26.5	35.6
DetectoRS [51]	CVPR 2021	ResNet-50	14.8	32.8	11.4	0.0	10.8	28.3	38.0
Cascade R-CNN w/RFLA [52]	AAAI 2025	ResNet-50+HS-FPN [15]	24.6	54.6	18.6	9.2	**25.0**	28.6	35.9
DNTR [53]	TGRS 2024	ResNet-50	18.2	45.5	10.9	7.0	18.5	22.3	30.0
one-stages detectors:									
FCOS [42]	ICCV 2019	ResNet-50	12.6	30.4	8.1	2.3	12.2	17.2	25.0
FCOS w/RFLA [52]	ECCV 2022	ResNet-50	15.4	36.3	10.9	6.0	17.6	18.5	20.7
DINO [32]	ICLR 2023	ResNet-50	23.2	56.6	15.4	9.9	23.1	29.3	37.6
ORFENet [7]	TGRS 2024	ResNet-50	16.6	38.3	11.6	5.9	17.6	29.9	23.0
KLDet [45]	TGRS 2024	ResNet-50	18.3	43.5	12.4	7.9	19.5	20.9	24.8
CFPT [54]	TGRS 2025	ResNet-50	7.6	22.8	3.1	1.5	6.9	11.5	22.7
Position-DETR [35]	TGRS 2025	ResNet-50	21.1	52.3	12.7	9.7	21.4	25.6	32.2
CSFPR-RTDETR [38]	TGRS 2025	CSFHNet	23.5	54.9	16.3	**10.3**	23.1	28.9	35.0
TGCADNet [55]	TCSVT 2026	CSPDarknet-s	**26.3**	**57.2**	**20.6**	-	-	-	-
YOLO11 [30]	CVPR 2024	C3k2Net-s	24.1	54.3	17.7	8.4	22.5	**32.3**	**42.7**
DCP-Net	-	-	**27.5**	**59.8**	**21.5**	**10.7**	**26.4**	**35.2**	**45.4**

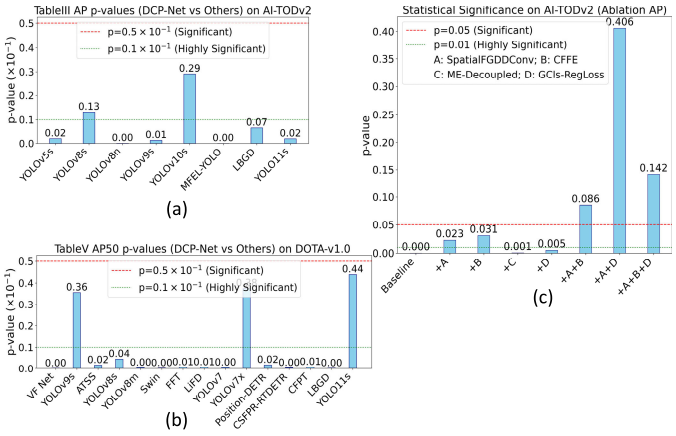


Fig. 7. Statistical significance analysis of comparative experiments and ablation experiments on the official dataset using Welch's t-test [50] (lower p-value indicates higher statistical significance).

B. Evaluation Metrics

We primarily use mean average precision (mAP), referred to as AP in the COCO dataset, as the evaluation metric. This includes basic metrics, AP (IoU = 0.5:0.95), AP50 (IoU = 0.5), and AP75 (IoU = 0.75), based on the IoU threshold, and scale metrics, APs, APm, and APl, based on the object pixel count. Due to the large number of tiny objects in the AI-TODv2 dataset, we use a more detailed AP scale division for it: very tiny (APvt), tiny (APt), small (APs), and medium (APm), corresponding to scale ranges of [2, 8], [8, 16], [16, 32], and [32, $+\infty$], respectively. The DOTA-v1.0 dataset is still evaluated using AP50. In addition, we use Giga floating-point operations (GFLOPs, GF), frames per second (FPS), and parameter size [Params, PA(M)] to evaluate model complexity and computational cost.

C. Implementation Details

All experiments were trained on a single RTX A6000 GPU, and the model code used Ultralytics YOLO11 as a baseline.

TABLE IV

PERFORMANCE COMPARISON OF DCP-NET AND OTHER SOTA YOLO DETECTORS ON THE AI-TODV2 TEST SET

Method	Venue	Resolution	AP	AP50
YOLOv5s [25]	CVPR 2022	640×640	22.52±0.87	51.36±1.23
YOLOv8s [27]	CVPR 2023	640×640	22.81±0.56	50.83±0.15
YOLOv8n [27]	CVPR 2023	640×640	17.70±0.29	41.92±0.41
YOLOv9s [28]	CVPR 2024	640×640	22.58±0.61	51.93±0.85
YOLOv10s [29]	CVPR 2024	640×640	**23.59±0.70**	51.11±0.47
MFEL-YOLO [8]	ESWA 2025	640×640	17.34±0.35	40.31±0.62
LBGD [37]	TCSVT 2026	640×640	18.04±0.74	43.61±1.27
YOLO11s [30]	CVPR 2024	640×640	23.36±0.42	**53.01±0.82**
DCP-Net	-	640×640	**25.52±0.27**	**57.34±0.32**

The environment was built using the PyTorch [49] framework. The optimizer was mainly SGD, with a weight decay of 0.0005 and a momentum of 0.937. More detailed information is provided in Table II, including image size, dataset partitioning, initial learning rate, training batches, epochs, and evaluation metrics required for validation. In all experiments, we did not use pretrained weights and trained the model from scratch.

D. Comparison With SOTA Methods

To evaluate the effectiveness of DCP-Net, we compare it with several SOTA detectors on two representative datasets, AI-TODv2 and DOTA-v1.0. In all comparison tables, the bolded values indicate the top one or two detection results under each performance metric. We conduct three independent repeated experiments, and all ablation and comparison experiments (except Table III, which reports only the best performance for illustration) are reported in the form of "mean ± standard deviation (std)." Meanwhile, we apply Welch's t-test [50] to assess statistical significance between DCP-Net and the comparison methods (including SOTA and ablation modules). Fig. 7 results show that most performance improvements satisfy $p < 0.05$, demonstrating the reliability and stability of the experiment results.

1) Results on AI-TODv2: In the quantitative results in Tables III and IV, we compare the performance of DCP-Net

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TABLE V
PERFORMANCE COMPARISON OF DCP-NET AND OTHER SOTA METHODS ON THE DOTA-V1.0 VAL SET

Method	Venue	AP50	APs50	APm50	FPS	PA(M)	GF
VF Net [56]	CVPR 2021	71.72±0.95	50.68±0.83	72.15±1.02	33.0	32.9	>1K
YOLOv9s [28]	CVPR 2024	76.51±0.41	61.39±0.66	76.48±0.25	185.1	7.29	26.8
ATSS [41]	CVPR 2020	70.81±0.87	51.62±0.79	71.89±0.41	35.0	32.2	>1K
YOLOv8s [27]	CVPR 2023	76.10±0.46	59.22±1.05	78.12±0.25	153.8	11.1	28.7
YOLOv8m [27]	CVPR 2023	72.21±0.75	53.45±1.86	73.64±1.13	142.0	25.8	124.0
Swin [57]	ICCV 2021	73.21±0.49	48.71±0.89	75.48±1.15	19.0	66.5	>1K
FFT [13]	CVPR 2022	70.43±0.85	51.61±2.77	70.34±0.93	20.0	43.5	>1K
LiFD [58]	CVPR 2020	68.22±0.79	45.68±0.82	69.21±0.90	11.0	42.3	>1K
YOLOv7 [26]	CVPR 2022	74.08±0.54	59.51±0.22	75.29±1.42	104.0	37.2	165.0
YOLOv7x [26]	CVPR 2022	75.62±0.83	**62.59±0.43**	62.78±0.75	67.0	74.7	296.0
Position-DETR [35]	TGRS 2025	74.91±0.46	59.12±0.94	**79.25±0.21**	41.2	49.9	184.0
CSFPR-RTDETR [38]	TGRS 2025	74.13±0.99	62.45±1.75	75.70±0.95	39.8	14.1	63.9
CFPT [54]	TGRS 2025	72.68±0.50	52.33±0.78	77.03±1.53	25.6	20.9	108.1
LBGD [37]	TCSVT 2026	70.32±0.47	50.84±0.89	73.68±0.35	217.3	3.01	8.1
YOLO11s [30]	CVPR 2024	**76.84±0.27**	56.91±0.90	78.72±1.10	196.0	10.7	26.4
DCP-Net	-	**78.03±0.22**	**62.89±1.00**	**80.42±0.84**	161.2	15.8	41.7

TABLE VI
ABLATION EXPERIMENTS FOR SINGLE MODULE AND MULTIPLE MODULES ON AI-TODV2 DATASET

SpatialFGDDConv	CFFE	ME-Decoupled Head	GCls-RegLoss	AP	AP50	AP75	APvt	APt	APs	APm
✗	✗	✗	✗	23.36±0.42	53.01±0.82	17.24±0.31	7.27±0.73	21.85±1.29	31.03±0.72	42.25±0.17
✓	✗	✗	✗	24.39±0.31	55.38±0.26	18.17±0.09	8.84±0.32	22.97±0.11	32.12±0.35	43.46±0.24
✗	✓	✗	✗	24.42±0.30	54.93±0.14	17.86±0.55	8.65±0.18	23.24±0.37	32.38±0.16	43.17±0.29
✗	✗	✓	✗	24.05±0.24	53.34±0.30	17.65±0.62	7.89±0.76	22.28±0.94	31.84±0.58	42.93±0.34
✗	✗	✗	✓	24.78±0.11	55.69±0.22	18.39±0.17	8.16±0.31	22.87±0.24	33.05±0.13	42.96±0.20
✓	✓	✗	✗	24.97±0.58	55.57±0.35	18.84±0.23	7.58±0.18	23.79±0.32	33.27±0.19	43.78±0.26
✓	✗	✗	✓	25.10±0.43	56.26±0.28	18.96±0.15	8.39±0.27	24.15±0.36	33.32±0.21	42.77±0.18
✓	✓	✗	✓	25.21±0.33	56.95±0.16	18.57±0.30	9.37±0.22	23.88±0.14	33.19±0.33	43.24±0.11
✓	✓	✓	✓	**25.52±0.27**(+2.16)	**57.34±0.32**(+4.33)	**19.92±0.28**(+2.68)	**10.93±0.19**(+3.66)	**24.47±0.25**(+2.62)	**33.56±0.32**(+2.53)	43.51±0.23

TABLE VII
PERFORMANCE IMPACT OF THE POSITION AND NUMBER OF SPATIALFGDDCONV. SFGDDC AND CS DENOTE SPATIALFGDDCONV AND C3K2-SFG. "W/SE" AND "WO/SE" DENOTE WITH AND WITHOUT THE LITESEATTENTION, MECHANISM, RESPECTIVELY

Model	N_{SFGDDC}	N_{CS}	AP	PA(M)	GF
YOLO11s	0	0	23.36±0.42	10.71	26.4
+SFGDDC wo/ SE	7	0	23.78±0.36	14.2	34.1
	6	0	23.84±0.22	14.2	33.9
+SFGDDC w/ SE	7	0	24.03±0.45	14.4	34.3
	6	0	24.27±0.19	14.4	34.1
	6	5	24.10±0.28	14.8	36.5
	6	2	**24.39±0.31**	14.4	35.0

with SOTA detectors on the AI-TODv2 test set. Table III demonstrates the performance of our approach alongside other SOTA detectors. It all exhibits strong competitiveness across seven evaluation metrics: AP, AP50, AP75, APvt, APt, APs, and APm. Compared with the baseline model YOLO11s, DCP-Net improved AP, AP50, and AP75 by 3.4%, 5.5%, and 3.8%, respectively. Table IV shows the comparison results between our DCP-Net and the SOTA YOLO detector. On the AI-TODv2 dataset, DCP-Net achieved the highest performance with 57.34% AP50 and 25.52% AP. It outperformed the second-best method by 4.33% in AP50 and by 1.93% in AP, highlighting the effectiveness of our innovative method. Likewise, Specifically, for detecting very tiny (vt), tiny (t), small (s), and medium (m) objects, our approach achieves AP improvements of 2.3%, 4.1%, 2.9%, and 2.7%, respectively. Furthermore, whether on two-stage detectors or one-stage detectors, especially compared with the latest proposed method, our method achieved the best results in all metrics, fully demonstrating the generalization performance of the method.

2) Results on DOTA-v1.0: Table V shows a performance comparison of different SOTA detectors on the DOTA-v1.0 dataset. Compared to frequency domain methods such as FFT and LiFD, DCP-Net reduces the number of parameters by approximately 27M and is also significantly superior to these two methods in terms of FPS and GFLOPs. It avoids expensive spectral transformation and achieves the best detection performance with an AP50 of 78.03%, which is 1.19% higher than the second-best YOLO11. In addition, compared with the second-best methods YOLOv7x and Position-DETR, APs50 and APm50 are 62.59% and 79.25%, respectively, representing improvements of 0.30% and 1.17%, where APs50 represents the AP index when detecting small(s) objects, and IoU is 0.50; APm50 is similar. Experimental results show that our method achieves the best balance between detection performance and training cost.

E. Ablation Study

In this section, we conducted a series of ablation experiments on the baseline model on the AI-TODv2 test set to demonstrate the effectiveness of our components. First, we evaluated the contributions of individual modules and combined modules to DCP-Net from the perspective of the overall model structure. Then, we conducted ablation experiments on these four modules, respectively, and verified the rationality of each module by replacing or removing some subcomponents.

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TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT FREQUENCY ENHANCEMENT METHODS

Baseline	SE	AdualDCDM	HighFreqProxy	AP	FPS
YOLO11s	✗	✗	✗	23.36±0.42	250.0
	✓	✓	FFT [13]	23.70±0.32	126.5
	✓	✓	FreqSelect [14]	23.93±0.45	129.8
	✓	✓	LaplaceConv	**24.39±0.31**	163.9

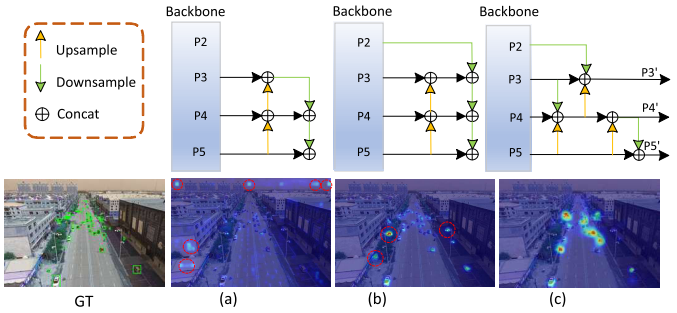


Fig. 8. Comparison of different FPN structures and their heatmaps on AI-TODv2 (red dashed circles indicate background noise). (a) Baseline PAFPN [59]. It focuses on most small objects but suffers from heavy background noise. (b) PAFPN+P2. It emphasizes small objects but lacks correct context guidance, failing to distinguish object scales. Nearby large objects are over-focused, causing false detections. (c) CFFE (ours). It not only covers small objects but also effectively detects slightly larger objects, achieving balanced recognition of objects of different scales while filtering out background noise.

1) Ablation Experiments of Each Modules: The ablation studies of DCP-Net are presented in Table VI. Compared with the baseline model, adding SpatialFGDDConv alone improves APvt, APt, and APs by 1.57%, 1.12%, and 1.09%, respectively, demonstrating its effectiveness in enhancing the feature for tiny objects. Introducing GCls-RegLoss alone improves AP by 1.42%, and the smaller standard deviation further indicates that it provides stable training gradients. Similarly, both CFFE and the ME-Decoupled Head individually achieve better detection performance. As more modules are progressively added, the detection performance (AP) consistently improves. Finally, when all modules are integrated into the baseline, the best performance is achieved, with AP, AP50, and AP75 improving by 2.16%, 4.33%, and 2.68%, respectively. Meanwhile, gains of 3.66% and 2.62% are obtained on tiny (t) and very tiny (vt) objects. These results not only validate the effectiveness of our method but also demonstrate that spatial-frequency perception and detail-context enhancement are effective strategies for improving remote sensing small object detection.

2) SpatialFGDDConv: As shown in Table VII, applying SpatialFGDDConv with LiteSEAttention in the last six layers effectively suppresses feature interference. Introducing two C3k2-SFG modules in shallow layers enables early capture of spatial-frequency features, significantly improving the perception of tiny objects (24.39% AP). In contrast, using SpatialFGDDConv at the initial convolution stage disrupts the input image's original spatial information, and excessive use of C3k2-SFG enlarges the receptive field, introducing irrelevant background noise and degrading detection performance.

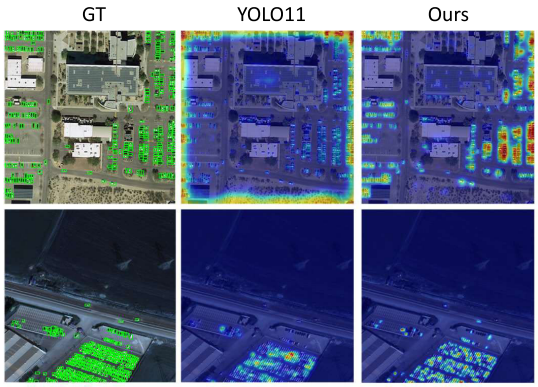


Fig. 9. Visualization of detection results and heatmaps on AI-TODv2. The highlighted areas represent regions of network interest.

TABLE IX

PERFORMANCE IMPACT OF DIFFERENT FEATURE FUSION STRUCTURES. "W/BILINEAR" DENOTES UPSAMPLING USING BILINEAR INTERPOLATION METHOD (ORIGINAL METHOD WAS NEAREST-NEIGHBOR INTERPOLATION METHOD)

Baseline	Neck	AP	FPS	PA(M)	GF
YOLO11s	PANFPN	23.36±0.42	250.0	10.71	26.4
	PANFPN+P2	24.13±0.28	212.8	10.87	28.5
	PANFPN w/Bilinear	23.79±0.45	232.3	9.41	21.4
	CFFE w/Bilinear	**24.42±0.30**	208.3	**9.95**	**23.9**

TABLE X

PERFORMANCE IMPACT OF VARIOUS CONVOLUTIONAL STRUCTURES FOR HEAD

Baseline	Reg Head	Cls Head	AP	FPS	PA(M)	GF
YOLO11s	Conv	Conv	23.36±0.42	250.0	10.7	26.4
	Conv	DWconv	23.26±0.13	277.6	9.4	21.3
	Conv	MidConv	20.30±0.08	263.3	12.4	35.8
	MidConv	MidConv	18.46±0.26	263.4	12.8	38.1
	MidConv	DWConv	22.42±0.22	270.2	11.1	28.7
	MidConv	Conv	**24.05±0.24**	**270.4**	**10.4**	**25.8**

Table VIII quantitatively demonstrates the superiority of our spatial-frequency perception method over existing frequency-domain approaches. Compared with FFT and FreqSelect, frequency-domain methods tend to amplify high-frequency noise due to their global transformations and also increase inference time. In contrast, the proposed LaplaceConv captures local high-frequency regions in the spatial domain, achieving the best performance (AP + 1.03%) with faster inference (FPS $\uparrow\approx$ 36). This shows that spatial-domain frequency approximation can more efficiently guide receptive field selection while preserving fine details of small objects.

3) CFFE: Table IX shows that CFFE achieves a +1.06% AP gain while significantly reducing GFLOPs and parameter count compared to the baseline. Although FPS slightly decreases, it remains acceptable for real-time remote sensing. In contrast, directly adding the P2 layer increases computational cost and complexity, and bilinear upsampling only partially alleviates feature discontinuity but fails to recover pixel-level details. By combining P2 with an optimized upsampling path, CFFE forms a more complete context transmission pipeline, achieving the best balance between accuracy and

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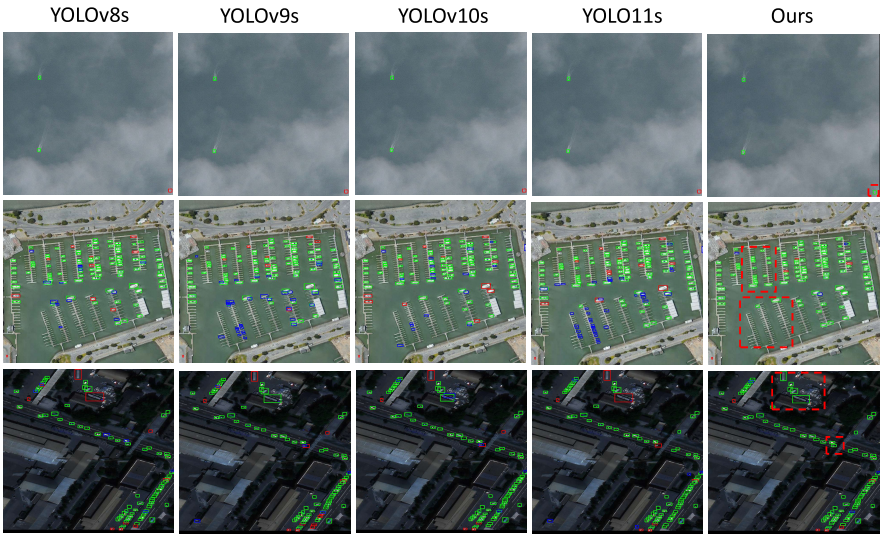


Fig. 10. Visualization comparison of DCP-Net detection results with other SOTA models on AI-TODv2. Green, blue, and red boxes represent true positive (TP), false positive (FP), and false negative (FN) predictions, respectively. Boxes with red dashed lines highlight areas of significant difference.



Fig. 11. Visualization comparison of DCP-Net with other SOTA models' detection results on DOTA-v1.0. Green, blue, and red boxes represent TP, FP, and FN predictions, respectively. Boxes with red dashed lines highlight areas of significant difference.

TABLE XI
PERFORMANCE IMPACT OF DIFFERENT LOSS FUNCTIONS. "W/NWD" DENOTES REGRESSION LOSS WITH NWD

Baseline	Cls Loss	Reg Loss	AP
✓	BCELoss	CIoULoss	23.36±0.42
✓	+FocalLoss	-	24.13±0.34
✓	-	+GRegLoss	24.48±0.20
✓	GClsLoss	-	24.43±0.29
✓	GClsLoss	+Attraction Loss w/GRegLoss	24.62±0.15
✓	GClsLoss	+Attraction Loss w/NWD [19]	24.02±0.33
✓	GClsLoss	+Attraction-Repulsion Loss	**24.78±0.11**

efficiency. Visualization in Fig. 8 further verifies its superiority in multiscale feature fusion.

4) ME-Decoupled Head: Table X shows that adding MidConv only to the regression branch in the ME-Decoupled Head effectively improves spatial alignment ability (especially scale and shape) and achieves +0.69% AP. This design efficiently reallocates computation, reaching best detection performance with lower cost (10.4M parameters, 25.8 GFLOPs) and higher speed (270 FPS). In contrast, applying MidConv or DWConv (depthwise separable convolution) in the classification branch both disrupts semantic discrimination and degrades performance. These results highlight the effectiveness of our head design in improving localization while maintaining efficiency.

5) GCls-RegLoss: Table XI presents the contributions of each component in GCls-RegLoss. Compared with the baseline, introducing focal loss alleviates sample imbalance (24.13% AP). The optimized GClsLoss demonstrates the advantage of adaptive sample assignment (24.43% AP). Moreover, using GRegLoss alone optimizes regression offset (24.48% AP) via distribution similarity (BC) and reduces gradient fluctuation (std 0.20). By combining GClsLoss with the attraction term of GRegLoss, further boosts AP to 24.62%, while using NWD as an attraction loss leads to gradient optimization conflicts due to the implicit strong cou-

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TABLE XII
COMPARISON OF SETTINGS FOR VARIOUS HYPERPARAMETERS IN THE LOSS FUNCTION

Baseline	a	α	Repulsion Loss	AP
✓	1	0	✗	23.36±0.42
✓	0.8	0.2	✓	**24.78±0.11**
✓	0.6	0.4	✓	24.54±0.16
✓	0.5	0.5	✓	24.41±0.13
✓	0.4	0.6	✓	24.24±0.22
✓	0.2	0.8	✓	24.33±0.31
✓	0	1	✓	24.17±0.27

pling between position and shape, resulting in a performance decrease (24.02% AP). Finally, incorporating the attraction–repulsion mechanism achieves the best performance (24.78% AP) and mitigates occlusion interference in dense scenes. Overall, the results verify that the proposed loss enhances sample learning and small-object regression.

Although GRegLoss provides stable gradients and precise alignment, it converges slowly when used alone. The simple computation of CIoU Loss compensates for this issue. Thus, we combine CIoU Loss with L_{Apg} (GRegLoss) to accelerate training while preserving their advantages. Experiments on AI-TODv2 determine optimal parameters in $Loss_{reg}$, as shown in Table XII. When $a = 0.8$ and $\alpha = 0.2$, the model achieves the best AP (24.78%) with efficient training.

F. Visualization Analysis

We demonstrate the visualization detection results of DCP-Net on two benchmark datasets, comparing them with baseline models and SOTA methods.

We visualize intuitively the heatmaps of the baseline model and our method on the AI-TODv2 test set in Fig. 9. The results show that the regions of interest focused by DCP-Net have higher consistency with the GT, especially in regions containing dense small objects, where the response is stronger. In contrast, the baseline model's heatmap exhibits weaker attention and more noise in these regions. This advantage stems from our enhanced ability to extract features from small objects and leverage surrounding features, thereby significantly suppressing noise and improving detection performance for dense small objects.

To further verify the performance superiority of our method, Figs. 10 and 11 present the qualitative comparison of the top-five SOTA methods on the AI-TODv2 and DOTA-v1.0 datasets, respectively. It can be seen that our method achieves superior detection performance in scenarios with complex backgrounds or large variations in object shape and scale. For example, when handling objects that are highly similar to the background (such as ships blending into clouds or harbors, and blurry vehicles under low-light conditions), other methods frequently suffer from false alarms or missed detections due to the lack of explicit distinction between object and background features. By contrast, DCP-Net separates object and background regions via SpatialFGDDConv, performs differentiated modeling of object details and contextual information, and remarkably reduces such detection errors. Moreover, such existing methods often exhibit severe bounding-box offsets when detecting objects with diverse scales or shapes (such as dense and multiscale vehicles, large ships, and elongated harbors), leading to substantial localization errors. By independently supervising stable regression of object position and shape, the loss function designed by us effectively improves localization accuracy.

These observations demonstrate that DCP-Net provides stronger feature discrimination and more robust localization for TOD in complex remote sensing scenes.

V. CONCLUSION

This article proposes DCP-Net for TOD in remote sensing images, specifically, integrating three plug-and-play modules and a novel loss function. To address the limitation of existing adaptive receptive field methods in complex backgrounds, we propose the SpatialFGDDConv module inspired by frequency-aware methods. It leverages spatial frequency features to distinguish foreground from background and, combined with dual dilated convolution, flexibly allocates receptive fields for both detail and context. To further reduce redundant noise, the CFFE module built upon SpatialFGDDConv restores lost small-object features in a cleaner and lightweight manner. Meanwhile, the ME-Decoupled Head improves regression accuracy for object scale and shape. In addition, GCls-RegLoss alleviates sample imbalance and gradient instability in existing losses, and effectively supervises detail–context learning. On AI-TODv2 and DOTA-v1.0, DCP-Net achieves 27.5% AP and 78.0% AP50, respectively, significantly outperforming existing methods. Although the FPS drops slightly due to computational overhead, the model still maintains competitive real-time performance. Currently, detection performance under extreme scenarios remains limited, and the weights of loss functions require adjustment based on different datasets, increasing experimental complexity. Future work will focus on improving the computational efficiency of the model while further enhancing its robustness and effectiveness in real-world remote sensing applications.

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